

Foundation (Jalapeno)

Q1. Solve the system: $x + y = 3$, $y = x^2 - 1$

- (A) (0, -1)
 (B) (1, 2)
 (C) (2, 1)
 (D) (1, 1), (-1, 3)
 (E) (2, 3)

Q2. Find the intersection point of $y = 2x - 3$ and $y = x^2 + 1$

- (A) (1, -1)
 (B) (1, 1), (-1, -3)
 (C) (-1, -5)
 (D) (2, 1)
 (E) (3, 3)

Q3. Solve the system: $y = 2x + 1$, $y = -x^2 + 2$

- (A) (-1, -1), (1, 3)
 (B) (0, 1), (2, 5)
 (C) (1, 3)
 (D) (2, 5)
 (E) (0, 0)

Q4. Which system has no real solution? $x + y = 2$, $y = x^2 + 3$

- (A) $y = x + 1$, $y = x - 3$
 (B) $x + y = 4$, $y = x - 2$
 (C) $x - y = 1$, $y = x + 2$
 (D) $y = x - 3$, $x + y = 0$
 (E) $x + y = 2$, $y = x^2 + 3$

Q5. Find the intersection points: $x - y = 1$, $y = x^2 - 2$

- (A) (1, 0), (-1, -2)
 (B) (-1, 0), (1, -2)
 (C) (1, 0), (2, 2)
 (D) (0, 0), (3, 7)
 (E) (-2, -3)

Q6. Solve the system: $y = 3x + 2$, $x = y^2 - 2$

- (A) (1, 2), (-2, -4)
 (B) (-2, -4)
 (C) (-1, -1)
 (D) (2, 4), (0, 0)
 (E) (-2, 2)

Q7. What is the intersection point of $x - y = 2$ and $y = x^2$?

- (A) (2, 0), (-1, 3)
 (B) (1, 0), (-2, -3)
 (C) (-2, -3), (2, 0)
 (D) (0, 2), (-2, -3)
 (E) (2, 2)

Q8. $y = x^2 - 1$, $x + y = 4$ intersection is

- (A) (2, 3), (0, 2), (-1, -2)
 (B) (0, 1), (3, 5), (-2, -3)
 (C) (-1, 0), (3, 4), (2, 0)
 (D) (0, 0), (-2, 0), (1, 0)
 (E) (1, 1), (-1, 0), (2, 0)

Open Ended Questions (FRQ)**Q9.** Solve the system $y = x + 2$ and $x = y^2 - 2$ and interpret the intersection point(s).**Q10.** For the system $x - 2y = -2$ and $y = x^2 - 1$, find and interpret all real solutions.

Chef's Tips

- Tip 1: Encourage students to use substitution and elimination methods
- Tip 2: Remind them to always check their solutions
- Tip 3: Suggest the use of graphing to visualize intersection points